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Analysis of Signals in Time Domain

- Basic signals
- Basic operations on signals
- Decomposition of complex signals to elementary signals

1. Basic Signals

➢ Basic Signals

Focus on the definition, characteristics and relationship between elementary signals;

Note that the distinction and connection between continuous signal and discrete signal

Continuous Time Signal

- Sinusoidal
- Real exponential
- Imaginary exponential
- Complex exponential
- Sampling
- Unit step
- Unit impulse
- Ramp
- …

Discrete Time Signal

- Sinusoidal
- Real exponential
- Imaginary exponential
- Complex exponential
- Unit impulse
- Unit step
- …

2. Basic operations on signals

➢ Basic operations on signals

Emphasis the description of mathematical concept and physical concept

Note that the correspondence and difference between continuous and discrete signals

Continuous

- reversal
- shifting
- extension
- compression
- addition
- multiplication
- differentiation
- integration
- …

Discrete

- reversal
- displacement
- interpolation
- compression
- addition
- multiplication
- difference
- summation
- …

3. Basic Decomposition

➢ Basic Decomposition

Signals can be broken down into different components

- DC component/AC component
- Odd/even components
- Real/imaginary quantities
- …

Signals can be represented as δ signals which are the main points of time domain analysis

- ☼ Continuous-time signals $x(t)$ can be expressed as a linear combination of $\delta(t)$
- ☼ Discrete-time signals $x[k]$ can be expressed as a linear combination of $\delta[k]$.

Time domain analysis of signals

Thinking:

Signals in engineering are generally not basic signals.

Why should we learn the analysis of basic signals?

Time domain analysis of signals

- Through the basic signal, basic operation and basic decomposition, the representation of complex signals can be realized, so as to transform the analysis of complex signals into the analysis of basic signals, which is the basic idea of signal analysis and processing.

- Basic signal and basic operations are also the basic carriers of signal frequency domain analysis and complex frequency domain analysis, which help us intuitively and clearly understand the correspondence between the signal time domain and the transformation domain and its characteristics.